8. Voltage Detection Circuit (LVDA)

The voltage detection circuit (LVDA) monitors the voltage level input to the VCC pin using a program.

8.1 Overview

For voltage detection 0, the detection voltage is selectable from among three different levels and the reset from voltage monitoring 0 can be enabled or disabled after a reset by using the option function select register 1 (OFS1).
For voltage detection 1 and voltage detection 2, the detection voltage is selectable from among three different levels by using the voltage detection level select register (LVDLVLR).
The reset from voltage monitoring 0, reset/interrupt from voltage monitoring 1, and reset/interrupt from voltage monitoring 2 can be used.

Table 8.1 lists the specifications of the voltage detection circuit. Figure 8.1 is a block diagram of the voltage detection circuit. Figure 8.2 is a block diagram of the voltage monitoring 1 interrupt/reset circuit. Figure 8.3 is a block diagram of the voltage monitoring 2 interrupt/reset circuit.

Table 8.1 Voltage Detection Circuit Specifications

Item		Voltage Monitoring 0	Voltage Monitoring 1	Voltage Monitoring 2
VCC monitoring	Monitored voltage	Vdet0	Vdet1	Vdet2
	Detection voltage	Selectable from among three different levels by using OFS1.VDSEL[1:0] bits	Selectable from among three different levels by using LVDLVLR.LVD1LVL[3:0] bits	Selectable from among three different levels by using LVDLVLR.LVD2LVL[3:0] bits
	Monitoring flag	None	LVD1SR.LVD1MON flag: Monitors whether voltage is higher or lower than Vdet1	LVD2SR.LVD2MON flag: Monitors whether voltage is higher or lower than Vdet2
			LVD1SR.LVD1DET flag: Vdet1 passage detection	LVD2SR.LVD2DET flag: Vdet2 passage detection
Process upon voltage detection	Reset	Voltage monitoring 0 reset	Voltage monitoring 1 reset	Voltage monitoring 2 reset
		Reset when Vdet0 > VCC CPU restart after specified time with VCC > Vdet0	Reset when Vdet1 > VCC CPU restart timing selectable: after specified time with VCC > Vdet1 or Vdet1 > VCC	Reset when Vdet2 > VCC CPU restart timing selectable: after specified time with VCC > Vdet2 or Vdet2 > VCC
	Interrupt	No interrupt	Voltage monitoring 1 interrupt	Voltage monitoring 2 interrupt
			Non-maskable interrupt or maskable interrupt selectable	Non-maskable interrupt or maskable interrupt selectable
			Interrupt request issued when Vdet1 > VCC and VCC > Vdet1 or either	Interrupt request issued when Vdet2 > VCC and VCC > Vdet2 or either
Digital filter	Enable/ Disable switching	Digital filter function not available	Available	Available
	Sampling time	—	1/n LOCO frequency × 2 (n: 2, 4, 8, 16)	1/n LOCO frequency × 2 (n: 2, 4, 8, 16)
Event linking		None	Available Output of event signals on detection of Vdet crossings	Available Output of event signals on detection of Vdet crossings

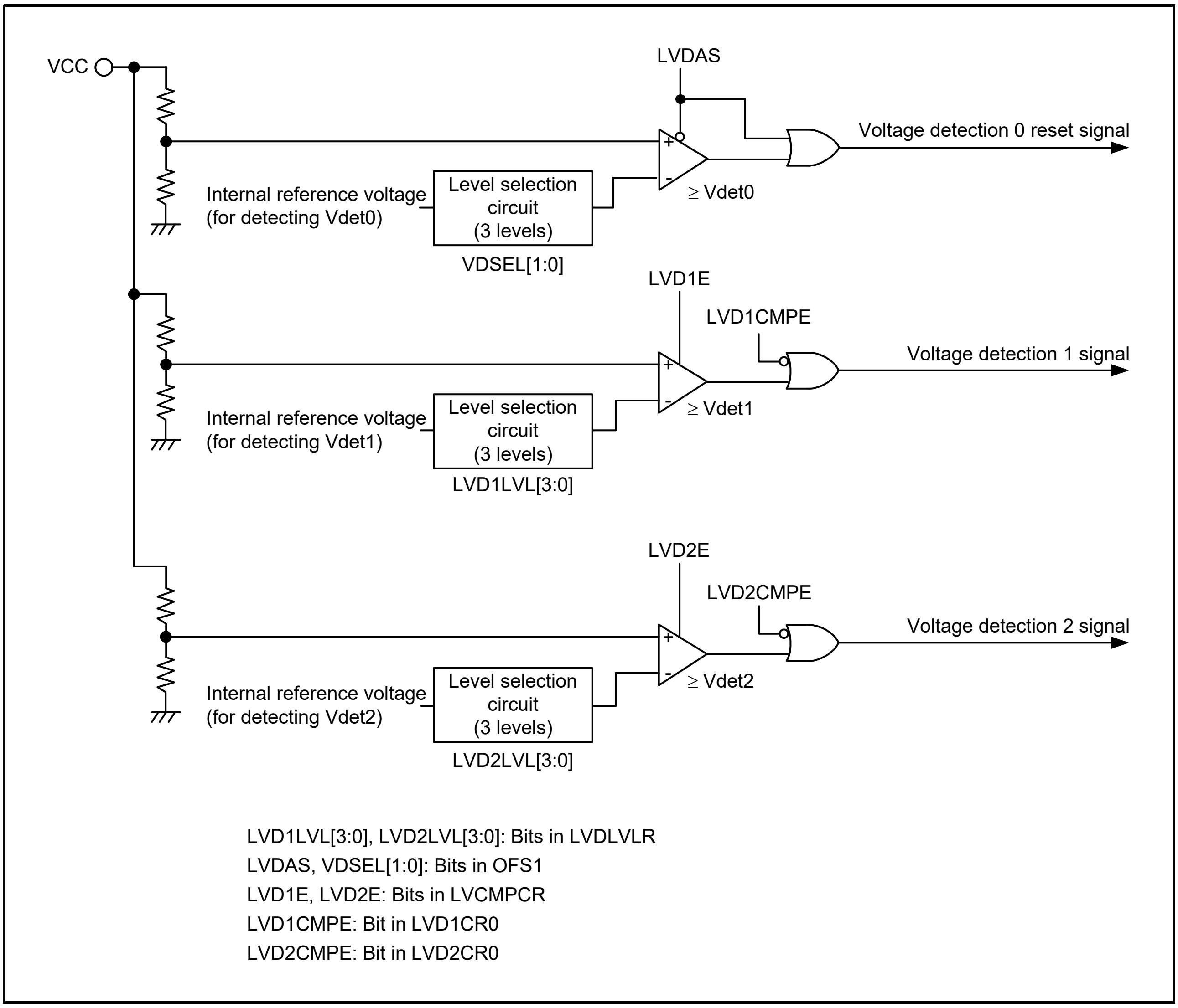
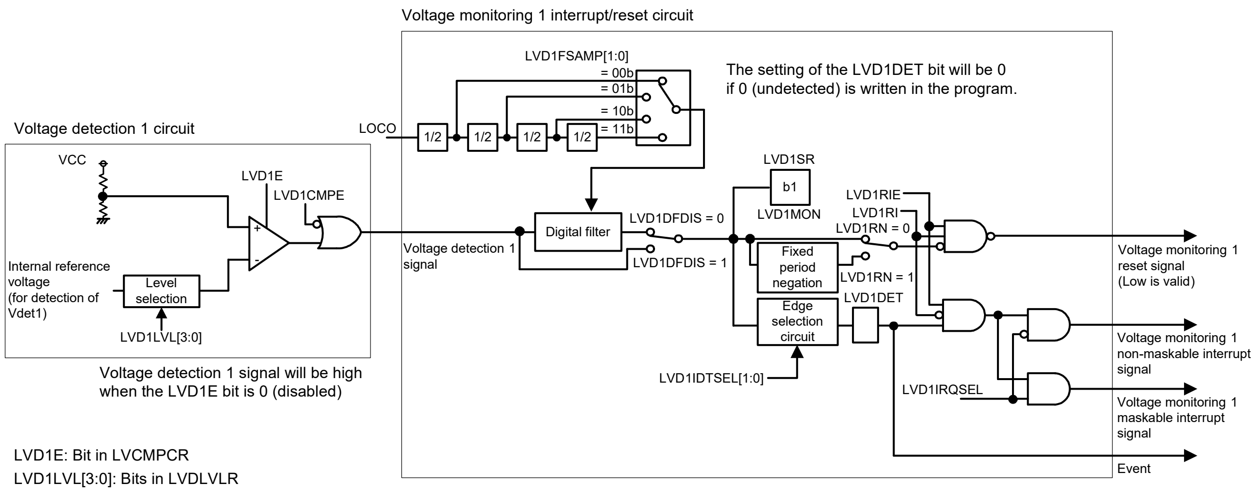
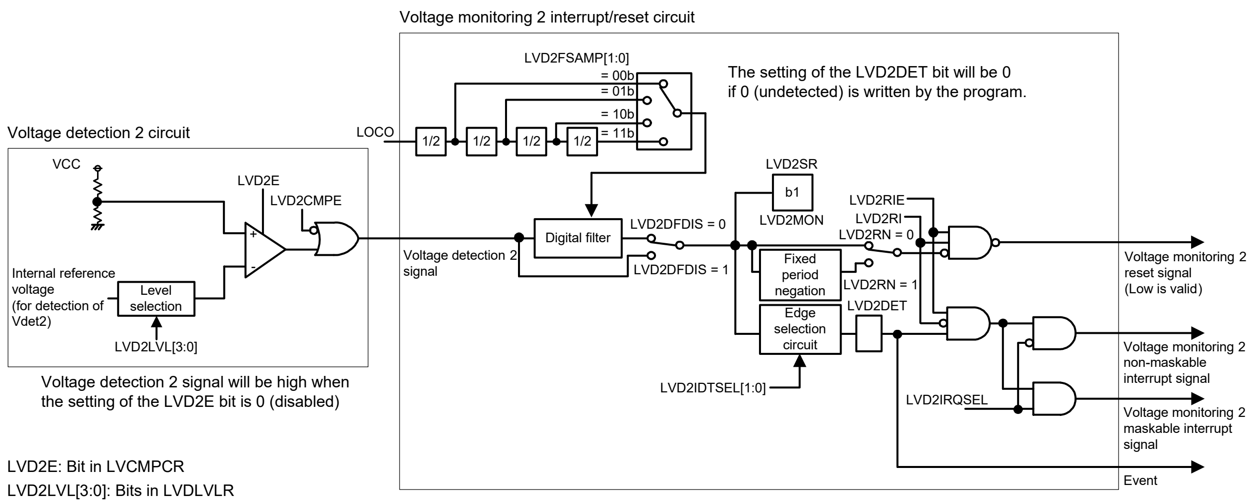


Figure 8.1 Block Diagram of Voltage Detection Circuit



LVD1E: Bit in LVCMPCR
LVD1LVL[3:0]: Bits in LVDLVLR
LVD1DFDIS, LVD1FSAMP[1:0], LVD1CMPE, LVD1RIE, LVD1RI, LVD1RN: Bits in LVD1CR0
LVD1IDTSEL[1:0], LVD1IRQSEL: Bits in LVD1CR1
LVD1DET, LVD1MON: Bits in LVD1SR

Figure 8.2 Block Diagram of Voltage Monitoring 1 Interrupt/Reset Circuit



LVD2E: Bit in LVCMPCR
LVD2LVL[3:0]: Bits in LVDLVLR
LVD2DFDIS, LVD2FSAMP[1:0], LVD2CMPE, LVD2RIE, LVD2RI, LVD2RN: Bits in LVD2CR0
LVD2IDTSEL[1:0], LVD2IRQSEL: Bits in LVD2CR1
LVD2DET, LVD2MON: Bits in LVD2SR

Figure 8.3 Block Diagram of Voltage Monitoring 2 Interrupt/Reset Circuit

8.2 Register Descriptions

8.2.1 Voltage Monitoring 1 Circuit Control Register 1 (LVD1CR1)

Address(es): 0008 00E0h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	LVD1IRQSEL	LVD1IDTSEL[1:0]	
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b1, b0	LVD1IDTSEL[1:0]	Voltage Monitoring 1 Interrupt Generation Condition Select	b1 b0 0 0: When VCC ≥ Vdet1 (rise) is detected 0 1: When VCC < Vdet1 (drop) is detected 1 0: When drop and rise are detected 1 1: Settings prohibited	R/W
b2	LVD1IRQSEL	Voltage Monitoring 1 Interrupt Type Select	0: Non-maskable interrupt 1: Maskable interrupt*1	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. When enabling maskable interrupts, do not change the value of the NMIER.LVD1EN bit on the ICU side from the reset state.

8.2.2 Voltage Monitoring 1 Circuit Status Register (LVD1SR)

Address(es): 0008 00E1h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	LVD1MON	LVD1DET
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	LVD1DET	Voltage Monitoring 1 Voltage Change Detection Flag	0: Not detected 1: Vdet1 passage detection	R(/W)*1
b1	LVD1MON	Voltage Monitoring 1 Signal Monitor Flag	0: VCC < Vdet1 1: VCC ≥ Vdet1 or LVD1MON is disabled	R
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to this bit. After writing 0 to this bit, it takes 2 system clock cycles for the bit to be read as 0.

LVD1DET Flag (Voltage Monitoring 1 Voltage Change Detection Flag)

The LVD1DET flag is enabled when the LVCMPCR.LVD1E bit is 1 (voltage detection 1 circuit enabled) and the LVD1CR0.LVD1CMPE bit is 1 (voltage monitoring 1 circuit comparison result output enabled).

The LVD1DET flag should be set to 0 after LVD1CR0.LVD1RIE is set to 0 (disabled). LVD1CR0.LVD1RIE can be set to 1 (enabled) after a period of two or more cycles of PCLKB has elapsed.

Depending on the number of cycles of PCLKB defined for access to read an I/O register, two or more cycles than PCLKB may have to be secured as waiting time.

LVD1MON Flag (Voltage Monitoring 1 Signal Monitor Flag)

The LVD1MON flag is enabled when the LVCMPCR.LVD1E bit is 1 (voltage detection 1 circuit enabled) and the LVD1CR0.LVD1CMPE bit is 1 (voltage monitoring 1 circuit comparison result output enabled).

8.2.3 Voltage Monitoring 2 Circuit Control Register 1 (LVD2CR1)

Address(es): 0008 00E2h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	LVD2IRQSEL	LVD2IDTSEL[1:0]	
Value after reset:	0	0	0	0	0	0	0	1

Bit	Symbol	Bit Name	Description	R/W
b1, b0	LVD2IDTSEL[1:0]	Voltage Monitoring 2 Interrupt Generation Condition Select	b1 b0 0 0: When VCC ≥ Vdet2 (rise) is detected 0 1: When VCC < Vdet2 (drop) is detected 1 0: When drop and rise are detected 1 1: Settings prohibited	R/W
b2	LVD2IRQSEL	Voltage Monitoring 2 Interrupt Type Select	0: Non-maskable interrupt 1: Maskable interrupt*1	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. When enabling maskable interrupts, do not change the value of the NMIER.LVD2EN bit on the ICU side from the reset state.

8.2.4 Voltage Monitoring 2 Circuit Status Register (LVD2SR)

Address(es): 0008 00E3h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	LVD2MON	LVD2DET
Value after reset:	0	0	0	0	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	LVD2DET	Voltage Monitoring 2 Voltage Change Detection Flag	0: Not detected 1: Vdet2 passage detection	R(/W)*1
b1	LVD2MON	Voltage Monitoring 2 Signal Monitor Flag	0: VCC < Vdet2 1: VCC ≥ Vdet2 or LVD2MON is disabled	R
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to this bit. After writing 0 to this bit, it takes 2 system clock cycles for the bit to be read as 0.

LVD2DET Flag (Voltage Monitoring 2 Voltage Change Detection Flag)

The LVD2DET flag is enabled when the LVCMPCR.LVD2E bit is 1 (voltage detection 2 circuit enabled) and the LVD2CR0.LVD2CMPE bit is 1 (voltage monitoring 2 circuit comparison result output enabled).
The LVD2DET flag should be set to 0 after LVD2CR0.LVD2RIE bit is set to 0 (disabled). LVD2CR0.LVD2RIE bit can be set to 1 (enabled) after a period of two or more cycles of PCLKB has elapsed.
Depending on the number of cycles of PCLKB defined for access to read an I/O register, two or more cycles of PCLKB may have to be secured as waiting time.

LVD2MON Flag (Voltage Monitoring 2 Signal Monitor Flag)

The LVD2MON flag is enabled when the LVCMPCR.LVD2E bit is 1 (voltage detection 2 circuit enabled) and the LVD2CR0.LVD2CMPE bit is 1 (voltage monitoring 2 circuit comparison result output enabled).

8.2.5 Voltage Monitoring Circuit Control Register (LVCMPCR)

Address(es): 0008 C297h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	LVD2E	LVD1E	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5	LVD1E	Voltage Detection 1 Enable*1	0: Voltage detection 1 circuit disabled 1: Voltage detection 1 circuit enabled	R/W
b6	LVD2E	Voltage Detection 2 Enable*2	0: Voltage detection 2 circuit disabled 1: Voltage detection 2 circuit enabled	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Note 1. The voltage of VCC = AVCC0 = AVCC1 when LVD1 is enabled must be set to at least 80 mV above the maximum value of the voltage detection 1 level selected by the LVDLVLR.LVD1LVL[3:0] bits.

Note 2. The voltage of VCC = AVCC0 = AVCC1 when LVD2 is enabled must be set to at least 80 mV above the maximum value of the voltage detection 2 level selected by the LVDLVLR.LVD2LVL[3:0] bits.

LVD1E Bit (Voltage Detection 1 Enable)

When using voltage detection 1 interrupt/reset or the LVD1SR.LVD1MON bit, set the LVD1E bit to 1. The voltage detection 1 circuit starts once td(E-A) passes after the LVD1E bit value is changed from 0 to 1. When using the voltage detection 1 circuit in deep software standby mode, do not set the DPSBYCR.DEEPCUT[1:0] bits to 11b.

LVD2E Bit (Voltage Detection 2 Enable)

When using voltage detection 2 interrupt/reset or the LVD2SR.LVD2MON bit, set the LVD2E bit to 1. The voltage detection 2 circuit starts once td(E-A) passes after the LVD2E bit value is changed from 0 to 1. When using the voltage detection 2 circuit in deep software standby mode, do not set the DPSBYCR.DEEPCUT[1:0] bits to 11b.

8.2.6 Voltage Detection Level Select Register (LVDLVLR)

Address(es): 0008 C298h

	b7	b6	b5	b4	b3	b2	b1	b0
	LVD2LVL[3:0]				LVD1LVL[3:0]			
Value after reset:	1	0	1	1	1	0	1	1

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	LVD1LVL[3:0]	Voltage Detection 1 Level Select (Standard voltage during drop in voltage)	b3 b0 1 0 0 1: 2.99 V (Vdet1_1) 1 0 1 0: 2.92 V (Vdet1_2) 1 0 1 1: 2.85 V (Vdet1_3) Settings other than above are prohibited.	R/W
b7 to b4	LVD2LVL[3:0]	Voltage Detection 2 Level Select (Standard voltage during drop in voltage)	b7 b4 1 0 0 1: 2.99 V (Vdet2_1) 1 0 1 0: 2.92 V (Vdet2_2) 1 0 1 1: 2.85 V (Vdet2_3) Settings other than above are prohibited.	R/W

The contents of the LVDLVLR register can only be changed if the LVCMPCR.LVD1E and LVCMPCR.LVD2E bits (voltage detection n circuit disable; n = 1, 2) are both 0. The voltage detection circuits 1 and 2 should not be set at the same voltage detection level.

8.2.7 Voltage Monitoring 1 Circuit Control Register 0 (LVD1CR0)

Address(es): 0008 C29Ah

	b7	b6	b5	b4	b3	b2	b1	b0
	LVD1RN	LVD1RI	LVD1FSAMP[1:0]		—	LVD1CMPE	LVD1DFDIS	LVD1RIE
Value after reset:	1	0	0	0	x	0	1	0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	LVD1RIE	Voltage Monitoring 1 Interrupt/Reset Enable	0: Disabled 1: Enabled	R/W
b1	LVD1DFDIS	Voltage Monitoring 1 Digital Filter Disable Mode Select	0: Digital filter enabled 1: Digital filter disabled	R/W
b2	LVD1CMPE	Voltage Monitoring 1 Circuit Comparison Result Output Enable	0: Voltage monitoring 1 circuit comparison result output disabled. 1: Voltage monitoring 1 circuit comparison result output enabled.	R/W
b3	—	Reserved	The read value is undefined. The write value should be 0.	R/W
b5, b4	LVD1FSAMP [1:0]	Sampling Clock Select	b5 b4 0 0: 1/2 LOCO frequency 0 1: 1/4 LOCO frequency 1 0: 1/8 LOCO frequency 1 1: 1/16 LOCO frequency	R/W
b6	LVD1RI	Voltage Monitoring 1 Circuit Mode Select	0: Voltage monitoring 1 interrupt during Vdet1 passage 1: Voltage monitoring 1 reset enabled when the voltage falls to and below Vdet1	R/W
b7	LVD1RN	Voltage Monitoring 1 Reset Negate Select	0: Negation follows a stabilization time (tLVD1) after VCC > Vdet1 is detected. 1: Negation follows a stabilization time (tLVD1) after assertion of the LVD1 reset.	R/W

LVD1RIE Bit (Voltage Monitoring 1 Interrupt/Reset Enable)

Ensure that neither a voltage monitoring 1 reset nor a voltage monitoring 1 interrupt is generated during programming or erasure of the flash memory.

LVD1DFDIS Bit (Voltage Monitoring 1 Digital Filter Disable Mode Select)

Set the LOCOCR.LCSTP bit to 0 (the LOCO operates) if the LVD1DFDIS bit is 0 (enabling the digital filter circuit). Set the LVD1DFDIS bit to 1 (digital filter circuit disabled) when using voltage monitoring 1 circuit in software standby mode or deep software standby mode.

LVD1FSAMP [1:0] Bits (Sampling Clock Select)

The LVD1FSAMP[1:0] bits can be modified only when the LVD1DFDIS bit is 1 (digital filter circuit disabled). The LVD1FSAMP[1:0] bits should not be modified when the LVD1DFDIS bit is 0 (digital filter circuit enabled).

LVD1RI Bit (Voltage Monitoring 1 Circuit Mode Select)

When the LVD1RI bit is 1 (voltage monitoring 1 reset selected) or when the LVD2CR0.LVD2RI bit is 1 (voltage monitoring 2 reset selected), a transition to deep software standby mode cannot be made, instead a transition to software standby mode is made. To enter deep software standby mode, set the LVD1RI bit to 0 (voltage monitoring 1 interrupt selected) and the LVD2CR0.LVD2RI bit to 0 (voltage monitoring 2 interrupt selected).

LVD1RN Bit (Voltage Monitoring 1 Reset Negate Select)

If the LVD1RN bit is to be set to 1 (negation follows a stabilization time after assertion of the LVD1 reset signal), set the LOCOCR.LCSTP bit to 0 (the LOCO operates). Furthermore, if a transition to software standby or deep software standby is to be made, the only possible value for the LVD1RN bit is 0 (negation follows a stabilization time after VCC > Vdet1 is detected). Do not set the LVD1RN bit to 1 (negation follows a stabilization time after assertion of the LVD1 reset signal) when this is the case.

8.2.8 Voltage Monitoring 2 Circuit Control Register 0 (LVD2CR0)

Address(es): 0008 C29Bh

	b7	b6	b5	b4	b3	b2	b1	b0
	LVD2RN	LVD2RI	LVD2FSAMP[1:0]		—	LVD2CMPE	LVD2DFDIS	LVD2RIE
Value after reset:	1	0	0	0	x	0	1	0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	LVD2RIE	Voltage Monitoring 2 Interrupt/Reset Enable	0: Disabled 1: Enabled	R/W
b1	LVD2DFDIS	Voltage Monitoring 2 Digital Filter Disable Mode Select	0: Digital filter enabled 1: Digital filter disabled	R/W
b2	LVD2CMPE	Voltage Monitoring 2 Circuit Comparison Result Output Enable	0: Voltage monitoring 2 circuit comparison result output disabled. 1: Voltage monitoring 2 circuit comparison result output enabled.	R/W
b3	—	Reserved	The read value is undefined. The write value should be 0.	R/W
b5, b4	LVD2FSAMP[1:0]	Sampling Clock Select	b5 b4 0 0: 1/2 LOCO frequency 0 1: 1/4 LOCO frequency 1 0: 1/8 LOCO frequency 1 1: 1/16 LOCO frequency	R/W
b6	LVD2RI	Voltage Monitoring 2 Circuit Mode Select	0: Voltage monitoring 2 interrupt during Vdet2 passage 1: Voltage monitoring 2 reset enabled when the voltage falls to and below Vdet2	R/W
b7	LVD2RN	Voltage Monitoring 2 Reset Negate Select	0: Negation follows a stabilization time (tLVD2) after VCC > Vdet2 is detected. 1: Negation follows a stabilization time (tLVD2) after assertion of the LVD2 reset.	R/W

LVD2RIE Bit (Voltage Monitoring 2 Interrupt/Reset Enable)

Ensure that neither a voltage monitoring 2 reset nor a voltage monitoring 2 interrupt is generated during programming or erasure of the flash memory.

LVD2DFDIS Bit (Voltage Monitoring 2 Digital Filter Disable Mode Select)

Set the LOCOCR.LCSTP bit to 0 (the LOCO operates) if the LVD1DFDIS bit is 0 (enabling the digital filter circuit). Set the LVD2DFDIS bit to 1 (digital filter circuit disabled) when using voltage monitoring 2 circuit in software standby mode or deep software standby mode.

LVD2FSAMP[1:0] Bits (Sampling Clock Select)

The LVD2FSAMP[1:0] bits can be modified only when the LVD2DFDIS bit is 1 (digital filter circuit disabled). The LVD2FSAMP[1:0] bits should not be modified when the LVD2DFDIS bit is 0 (digital filter circuit enabled).

LVD2RI Bit (Voltage Monitoring 2 Circuit Mode Select)

When the LVD2RI it is 1 (voltage monitoring 2 reset selected) or when the LVD1CR0.LVD1RI bit is 1 (voltage monitoring 1 reset selected), a transition to deep software standby mode cannot be made, instead a transition to software standby mode is made. To enter to deep software standby mode, set the LVD2RI bit to 0 (voltage monitoring 2 interrupt selected) and the LVD1CR0.LVD1RI bit to 0 (voltage monitoring 1 interrupt selected).

LVD2RN Bit (Voltage Monitoring 2 Reset Negate Select)

If the LVD2RN bit is to be set to 1 (negation follows a stabilization time after assertion of the LVD2 reset signal), set the LOCOCR.LCSTP bit to 0 (the LOCO operates). Furthermore, if a transition to software standby or deep software standby is to be made, the only possible value for the LVD2RN bit is 0 (negation follows a stabilization time after VCC > Vdet2 is detected). Do not set the LVD2RN bit to 1 (negation follows a stabilization time after assertion of the LVD2 reset signal) when this is the case.

8.3 VCC Input Voltage Monitor

8.3.1 Monitoring Vdet0

Monitoring Vdet0 is not possible.

8.3.2 Monitoring Vdet1

Table 8.2 lists the procedures for setting up monitoring against Vdet1. After the settings are completed, results of comparison by voltage monitoring 1 can be monitored by using the LVD1SR.LVD1MON flag.

Table 8.2 Procedures for Setting up Monitoring against Vdet1

Step		Monitoring the Results of Comparison by Voltage Monitoring 1
Setting the voltage detection 1 circuit	1	Select the detection voltage by setting the LVDLVLR.LVD1LVL[3:0] bits.
	2	Set LVCMPCR.LVD1E = 1 (enabling the voltage detection 1 circuit).
	3	Wait for at least td(E-A) (LVD operation stabilization time after LVD is enabled).*1
Setting the digital filter *2	4	Select the sampling clock for the digital filter by setting the LVD1CR0.LVD1FSAMP[1:0] bits.
	5	Set LVD1CR0.LVD1DFDIS = 0 (enabling the digital filter).
	6	Wait for at least 2n + 3 cycles of the LOCO (where n = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by n).
Enabling output	7	Set LVD1CR0.LVD1CMPE = 1 (enabling output of the results of comparison by voltage monitoring 1).

Note 1. Steps 4 to 6 can be performed during the waiting time of step 3. For details of td(E-A), refer to section 63, Electrical Characteristics.

Note 2. Steps 4 to 6 are not required if the digital filter is not in use.

8.3.3 Monitoring Vdet2

Table 8.3 lists the procedures for setting up monitoring against Vdet2. After the settings are completed, results of comparison by voltage monitoring 2 can be monitored by using the LVD2SR.LVD2MON flag.

Table 8.3 Procedures for Setting up Monitoring against Vdet2

Step		Monitoring the Results of Comparison by Voltage Monitoring 2
Setting the voltage detection 2 circuit	1	Select the detection voltage by setting the LVDLVLR.LVD2LVL[3:0] bits.
	2	Set LVCMPCR.LVD2E = 1 (enabling the voltage detection 2 circuit).
	3	Wait for at least td(E-A) (LVD operation stabilization time after LVD is enabled).*1
Setting the digital filter *2	4	Select the sampling clock for the digital filter by setting the LVD2CR0.LVD2FSAMP[1:0] bits.
	5	Set LVD2CR0.LVD2DFDIS = 0 (enabling the digital filter).
	6	Wait for at least 2n + 3 cycles of the LOCO (where n = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by n).
Enabling output	7	Set LVD2CR0.LVD2CMPE = 1 (enabling output of the results of comparison by voltage monitoring 2).

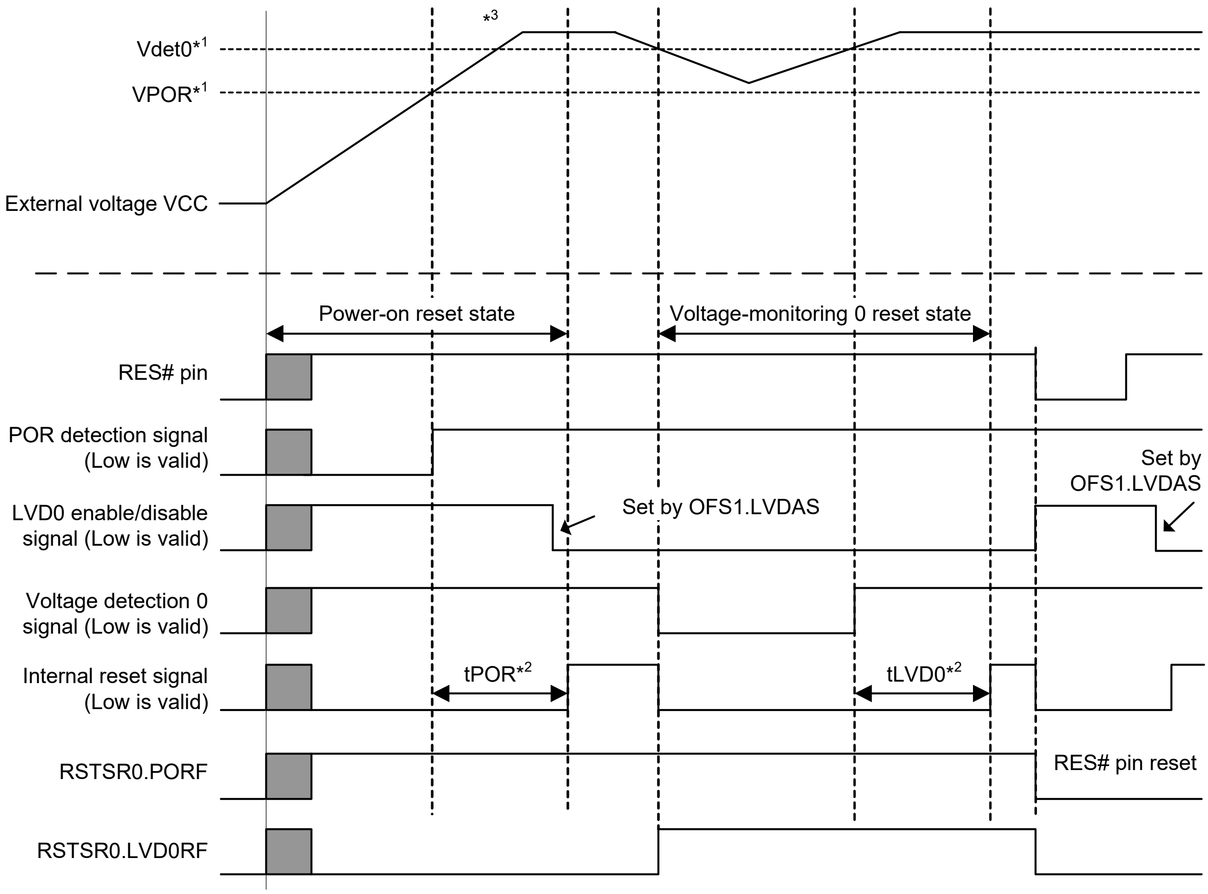
Note 1. Steps 4 to 6 can be performed during the waiting time of step 3. For details of td(E-A), refer to section 63, Electrical Characteristics.

Note 2. Steps 4 to 6 are not required if the digital filter is not in use.

8.4 Reset from Voltage Monitor 0

When using the reset from voltage monitor 0, clear the OFS1.LVDAS bit to 0 (enabling the voltage monitor 0 reset after a reset).

Figure 8.4 shows an example of operations for a voltage monitoring 0 reset.



Note: For details of the electrical characteristics, refer to section 63, Electrical Characteristics.

Note 1. VPOR indicates the detection level for a power-on reset and Vdet0 indicates the detection level for a voltage monitor 0 reset.

Note 2. tPOR indicates the period of a power-on reset and tLVD0 indicates the period of a voltage monitor 0 reset.

Note 3. At the time the power-supply voltage rises, VCC must rise to at least the minimum guaranteed voltage before release from the POR reset state.

Figure 8.4 Example of Voltage Monitoring 0 Reset Operation

8.5 Interrupt and Reset from Voltage Monitor 1

An interrupt or reset can be generated in response to the results of comparison by the voltage detection 1 circuit. Table 8.4 lists the procedures for setting bits related to the voltage monitor 1 interrupt and voltage monitor 1 reset so that voltage monitoring operates. Table 8.5 lists the procedure for setting bits related to the voltage monitor 1 interrupt and voltage monitor 1 reset so that voltage monitoring stops. Figure 8.5 shows an example of operations for a voltage monitor 1 interrupt. For the operation of the voltage monitor 1 reset, refer to Figure 6.2 in section 6, Resets.
Furthermore, if you intend to use the voltage monitoring 1 circuit in software standby or deep software standby mode, make settings for the voltage monitoring 1 circuit according to the following procedures.

(1) Setting in software standby mode
- Disable the digital filter (LVD1DFDIS = 1).
- After VCC > Vdet1 is detected, negate the voltage monitoring 1 reset signal (LVD1RN = 0) following a stabilization time.

(2) Settings in deep software standby mode
- Disable the digital filter (LVD1DFDIS = 1).
- Enable voltage monitoring 1 interrupts (LVD1RI = 0). If the voltage monitoring 1 reset is enabled (LVD1RI = 1), a transition to deep software standby mode will not be possible, and the transition will be to software standby mode instead.
- When the DPSBYCR.DEEPCUT[1:0] bits are 11b, the voltage monitoring 1 circuit is stopped. If you intend to use the voltage monitoring 1 circuit in deep software standby mode, set the DPSBYCR.DEEPCUT[1:0] bits to a value other than 11b.

Table 8.4 Procedures for Setting Bits Related to the Voltage Monitoring 1 Interrupt and Voltage Monitoring 1 Reset so that Voltage Monitoring Operates

<table>
<tr><th colspan="2">Step</th><th>Voltage Monitoring 1 Interrupt
(Voltage Monitoring 1 ELC Event Output)</th><th>Voltage Monitoring 1 Reset</th></tr>
<tr><td rowspan="3">Setting the voltage detection 1 circuit</td><td>1</td><td colspan="2">Select the detection voltage by setting the LVDLVLR.LVD1LVL[3:0] bits.</td></tr>
<tr><td>2</td><td colspan="2">Set LVCMPCR.LVD1E = 1 (enabling the voltage detection 1 circuit).*4</td></tr>
<tr><td>3</td><td colspan="2">Wait for at least td(E-A) (LVD operation stabilization time after LVD is enabled).*1</td></tr>
<tr><td rowspan="3">Setting the digital filter *2</td><td>4</td><td colspan="2">Select the sampling clock for the digital filter by setting the LVD1CR0.LVD1FSAMP[1:0] bits.</td></tr>
<tr><td>5</td><td colspan="2">Set LVD1CR0.LVD1DFDIS = 0 (enabling the digital filter).</td></tr>
<tr><td>6</td><td colspan="2">Wait for at least 2n + 3 cycles of the LOCO
(where n = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by n).</td></tr>
<tr><td rowspan="2">Setting the voltage monitoring 1 interrupt or reset</td><td>7</td><td>Set LVD1CR0.LVD1RI = 0 (selecting the voltage monitoring 1 interrupt).</td><td>• Set LVD1CR0.LVD1RI = 1 (selecting the voltage monitoring 1 reset).
• Select the type of the reset negation by setting the LVD1CR0.LVD1RN bit.</td></tr>
<tr><td>8</td><td>• Select the timing of interrupt requests by setting the LVD1CR1.LVD1IDTSEL[1:0] bits.
• Select the type of interrupt by setting the LVD1CR1.LVD1IRQSEL bit.</td><td>—</td></tr>
<tr><td rowspan="3">Enabling output</td><td>9</td><td colspan="2">Set LVD1SR.LVD1DET = 0.</td></tr>
<tr><td>10</td><td colspan="2">Set LVD1CR0.LVD1RIE = 1 (enabling the voltage monitoring 1 interrupt or reset).*3</td></tr>
<tr><td>11</td><td colspan="2">Set LVD1CR0.LVD1CMPE = 1 (enabling output of the results of comparison by voltage monitoring 1).</td></tr>
</table>

Note 1. Steps 4 to 10 can be performed during the waiting time of step 3. For details of td(E-A), refer to section 63, Electrical Characteristics.
Note 2. Steps 4 to 6 are not required if the digital filter is not in use.
Note 3. Step 10 is not required if only the ELC event signal is to be output.
Note 4. The voltage of VCC = AVCC0 = AVCC1 when LVD1 is enabled must be set to at least 80 mV above the maximum value of the voltage detection 1 level selected by the LVDLVLR.LVD1LVL[3:0] bits.

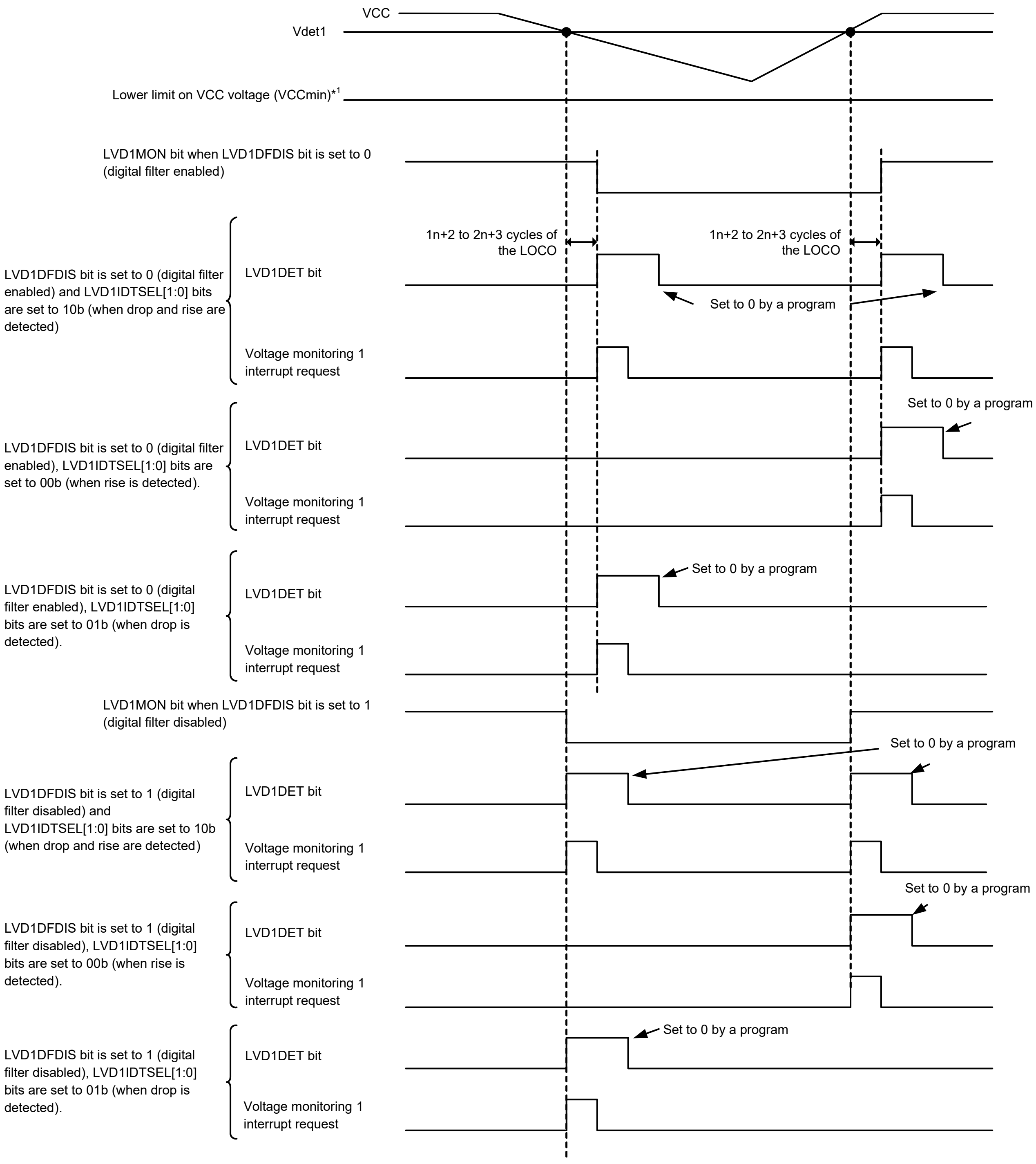
Table 8.5 Procedures for Setting Bits Related to the Voltage Monitoring 1 Interrupt and Voltage Monitoring 1 Reset so that Voltage Monitoring Stops

Step		Voltage Monitoring 1 Interrupt (Voltage Monitoring 1 ELC Event Output), Voltage Monitoring 1 Reset
Settings to stop enabling of output	1	Set LVD1CR0.LVD1CMPE = 0 (disabling output of the results of comparison by voltage monitoring 1).
	2	Wait for at least 2n + 3 cycles of the LOCO (where n = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by n).*1
	3	Set LVD1CR0.LVD1RIE = 0 (disabling the voltage monitoring 1 interrupt or reset).*2
Stopping the digital filter	4	Set LVD1CR0.LVD1DFDIS = 1 (disabling the digital filter).*1, *3
Stopping the voltage detection 1 circuit	5	Set LVCMPCR.LVD1E = 0 (disabling the voltage detection 1 circuit).

Note 1. Steps 2 and 4 are not required if the digital filter is not in use.
Note 2. Step 3 is not required if only the ELC event signal is to be output.
Note 3. To disable the digital filter from its enabled state and then re-enable it, disable it and wait for at least two cycles of the LOCO before re-enabling it.

If the voltage monitoring 1 interrupt or voltage monitoring 1 reset setting is to be made again after it has been used and stopped once, the following steps in the procedures for stopping and making the setting can be omitted according to the condition.

- Setting or stopping the voltage detection 1 circuit is not required if the setting for the voltage detection 1 circuit is not to be changed.
- Setting or stopping the digital filter is not required if the setting for the digital filter is not to be changed.
- Setting the voltage monitoring 1 interrupt or reset is not required if the setting for the voltage monitoring 1 interrupt or voltage monitoring 1 reset is not to be changed.



n: the frequency of the sampling clock for the digital filter is the LOCO frequency divided by n

Note 1. When the voltage monitoring 0 reset is not in use, VCC ≥ VCCmin.

Figure 8.5 Example of Voltage Monitoring 1 Interrupt Operation

8.6 Interrupt and Reset from Voltage Monitor 2

An interrupt or reset can be generated in response to the results of comparison by the voltage detection 2 circuit. Table 8.6 lists the procedures for setting bits related to the voltage monitor 2 interrupt and voltage monitor 2 reset so that voltage monitoring operates.Table 8.7 lists the procedure for setting bits related to the voltage monitor 2 interrupt and voltage monitor 2 reset so that voltage monitoring stops. Figure 8.6 shows an example of operations for a voltage monitor 2 interrupt. For the operation of the voltage monitor 2 reset, refer to Figure 6.2 in section 6, Resets. Furthermore, if you intend to use the voltage monitoring 2 circuit in software standby or deep software standby mode, make settings for the voltage monitoring 2 circuit according to the following procedures.

(1) Setting in software standby mode

- Disable the digital filter (LVD2DFDIS = 1).
- After VCC > Vdet2 is detected, negate the voltage monitoring 2 reset signal (LVD2RN = 0) following a stabilization time.

(2) Settings in deep software standby mode

- Disable the digital filter (LVD2DFDIS = 1).
- Enable voltage monitoring 2 interrupts (LVD2RI = 0). If the voltage monitoring 2 reset is enabled (LVD2RI = 1), a transition to deep software standby mode will not be possible, and the transition will be to software standby mode instead.
- When the DPSBYCR.DEEPCUT[1:0] bits are 11b, the voltage monitoring 2 circuit is stopped. If you intend to use the voltage monitoring 2 circuit in deep software standby mode, set the DPSBYCR.DEEPCUT[1:0] bits to a value other than 11b.

Table 8.6 Procedures for Setting Bits Related to the Voltage Monitoring 2 Interrupt and Voltage Monitoring 2 Reset so that Voltage Monitoring Operates

<table>
<tr><th colspan="2">Step</th><th>Voltage Monitoring 2 Interrupt
(Voltage Monitoring 2 ELC Event Output)</th><th>Voltage Monitoring 2 Reset</th></tr>
<tr><td rowspan="3">Setting the voltage detection 2 circuit</td><td>1</td><td colspan="2">Select the detection voltage by setting the LVDLVLR.LVD2LVL[3:0] bits.</td></tr>
<tr><td>2</td><td colspan="2">Set LVCMPCR.LVD2E = 1 (enabling the voltage detection 2 circuit).*4</td></tr>
<tr><td>3</td><td colspan="2">Wait for at least td(E-A) (LVD operation stabilization time after LVD is enabled).*1</td></tr>
<tr><td rowspan="3">Setting the digital filter *2</td><td>4</td><td colspan="2">Select the sampling clock for the digital filter by setting the LVD2CR0.LVD2FSAMP[1:0] bits.</td></tr>
<tr><td>5</td><td colspan="2">Set LVD2CR0.LVD2DFDIS = 0 (enabling the digital filter).</td></tr>
<tr><td>6</td><td colspan="2">Wait for at least 2n + 3 cycles of the LOCO
(where n = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by n).</td></tr>
<tr><td rowspan="2">Setting the voltage monitoring 2 interrupt or reset</td><td>7</td><td>Set LVD2CR0.LVD2RI = 0 (selecting the voltage monitoring 2 interrupt).</td><td>• Set LVD2CR0.LVD2RI = 1 (selecting the voltage monitoring 2 reset).
• Select the type of the reset negation by setting the LVD2CR0.LVD2RN bit.</td></tr>
<tr><td>8</td><td>• Select the timing of interrupt requests by setting the LVD2CR1.LVD2IDTSEL[1:0] bits.
• Select the type of interrupt by setting the LVD2CR1.LVD2IRQSEL bit.</td><td>—</td></tr>
<tr><td rowspan="3">Enabling output</td><td>9</td><td colspan="2">Set LVD2SR.LVD2DET = 0.</td></tr>
<tr><td>10</td><td colspan="2">Set LVD2CR0.LVD2RIE = 1 (enabling the voltage monitoring 2 interrupt or reset).*3</td></tr>
<tr><td>11</td><td colspan="2">Set LVD2CR0.LVD2CMPE = 1 (enabling output of the results of comparison by voltage monitoring 2).</td></tr>
</table>

Note 1. Steps 4 to 10 can be performed during the waiting time of step 3. For details of td(E-A), refer to section 63, Electrical Characteristics.

Note 2. Steps 4 to 6 are not required if the digital filter is not in use.

Note 3. Step 10 is not required if only the ELC event signal is to be output.

Note 4. The voltage of VCC = AVCC0 = AVCC1 when LVD2 is enabled must be set to at least 80 mV above the maximum value of the voltage detection 2 level selected by the LVDLVLR.LVD2LVL[3:0] bits.

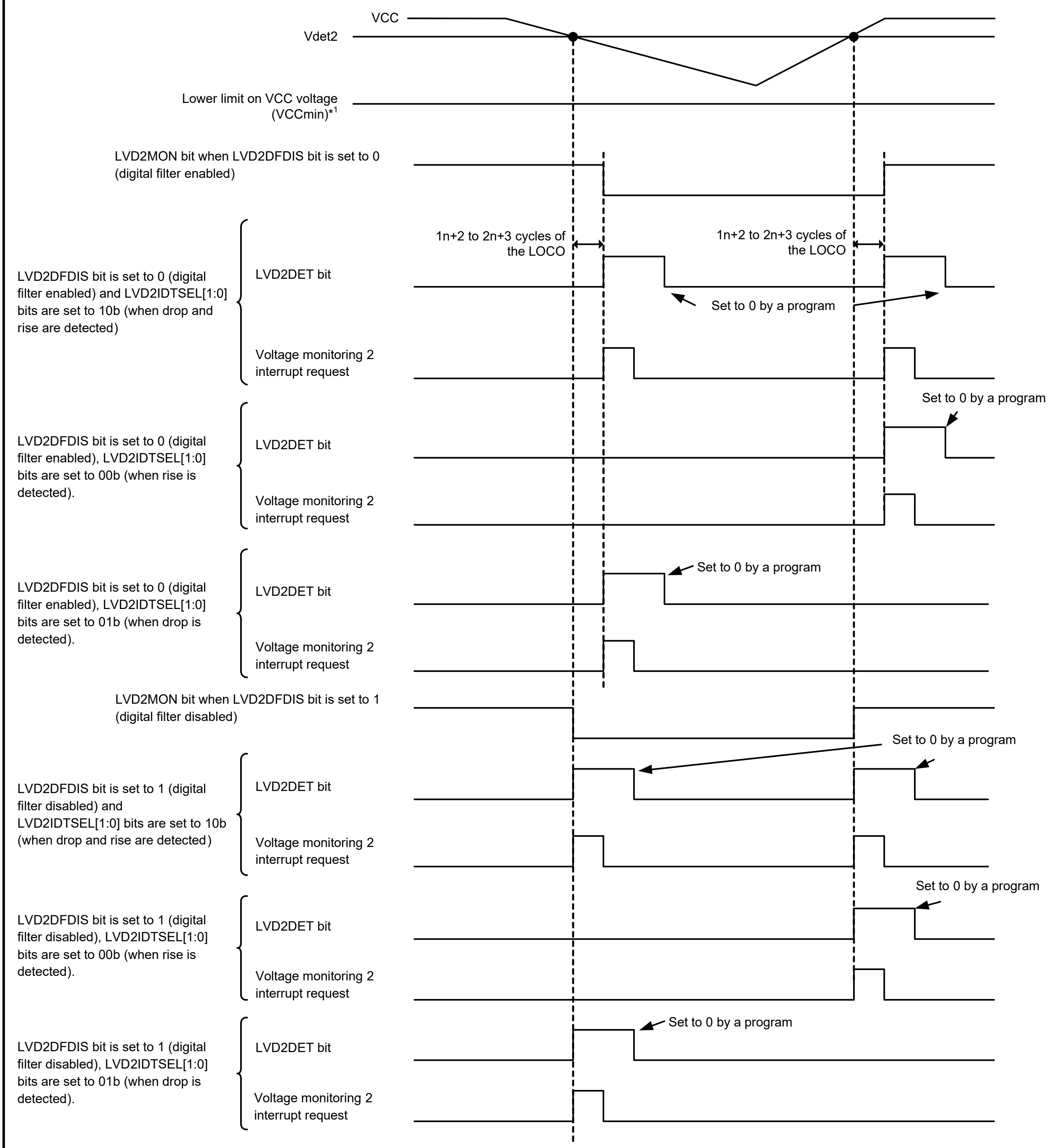
Table 8.7 Procedures for Setting Bits Related to the Voltage Monitoring 2 Interrupt and Voltage Monitoring 2 Reset so that Voltage Monitoring Stops

Step		Voltage Monitoring 2 Interrupt (Voltage Monitoring 2 ELC Event Output), Voltage Monitoring 2 Reset
Settings to stop enabling of output	1	Set LVD2CR0.LVD2CMPE = 0 (disabling output of the results of comparison by voltage monitoring 2).
	2	Wait for at least 2n + 3 cycles of the LOCO (where n = 2, 4, 8, 16, and the sampling clock for the digital filter is the LOCO frequency-divided by n).*1
	3	Set LVD2CR0.LVD2RIE = 0 (disabling the voltage monitoring 2 interrupt or reset).*2
Stopping the digital filter	4	Set LVD2CR0.LVD2DFDIS = 1 (disabling the digital filter).*1, *3
Stopping the voltage detection 2 circuit	5	Set LVCMPCR.LVD2E = 0 (disabling the voltage detection 2 circuit).

Note 1. Steps 2 and 4 are not required if the digital filter is not in use.
Note 2. Step 3 is not required if only the ELC event signal is to be output.
Note 3. To disable the digital filter from its enabled state and then re-enable it, disable it and wait for at least two cycles of the LOCO before re-enabling it.

If the voltage monitoring 2 interrupt or voltage monitoring 2 reset setting is to be made again after it has been used and stopped once, the following steps in the procedures for stopping and making the setting can be omitted according to the condition.

- Setting or stopping the voltage detection 2 circuit is not required if the setting for the voltage detection 2 circuit is not to be changed.
- Setting or stopping the digital filter is not required if the setting for the digital filter is not to be changed.
- Setting the voltage monitoring 2 interrupt or reset is not required if the setting for the voltage monitoring 2 interrupt or voltage monitoring 2 reset is not to be changed.



n: the frequency of the sampling clock for the digital filter is the LOCO frequency divided by n

Note 1. When the voltage monitoring 0 reset is not in use, VCC ≥ VCCmin.

Figure 8.6 Example of Voltage Monitoring 2 Interrupt Operation

8.7 Event Link Output

The LVD can output the event signals to the event link controller (ELC).

(1) Vdet1 passage detection event

The LVD outputs the event signal when it is detected that the voltage has passed the Vdet1 voltage while both the voltage detection 1 circuit and the voltage monitoring l circuit comparison result output are enabled.

(2) Vdet2 passage detection event

The LVD outputs the event signal when it is detected that the voltage has passed the Vdet2 voltage while both the voltage detection 2 circuit and the voltage monitoring 2 circuit comparison result output are enabled.

When enabling the LVD's event link output function, be sure to make settings for enabling the LVD before enabling the LVD event link function of the ELC. To stop the LVD's event link output function, be sure to make settings for stopping the LVD after disabling the LVD event link function of the ELC.

8.7.1 Interrupt Handling and Event Linking

The LVD has the bits to separately enable or disable the voltage monitoring l and 2 interrupts. When an interrupt source is generated and the interrupt is enabled by the interrupt enable bit, the interrupt signal (LVD1RIE and LVD2RIE) is output to the CPU.
On the contrary, as soon as an interrupt source is generated, the event link signal is output as the event signal to the other module via the ELC regardless of the state of the interrupt enable bit.
It is possible to output voltage monitoring l and 2 interrupts in software standby and deep software standby modes. The event signals for the ELC in software standby and deep software standby modes, are output as follows:

- When the event Vdet1/Vdet2 are detected in software standby mode, no event signals are generated for the ELC because no clock is presented in software standby mode. Since Vdet1/Vdet2 passage detection flags are preserved, however, when the supply of the clock is resumed after restoring from software standby mode, the event signals for the ELC are output according to the state of the Vdet1/Vdet2 passage detection flags.
- If events of passing Vdet1/Vdet2 are detected in deep software standby mode, no event signals are generated for the ELC.